Telco Customer Churn Prediction using Logistic Regression

1. Introduction

Customer churn refers to customers discontinuing a service and switching to competitors. Predicting customer churn is important because acquiring new customers is generally more expensive than retaining existing ones. This project aims to build a Logistic Regression model capable of predicting customer churn using customer demographic, account, and service-related information.

Dataset Source:
https://www.kaggle.com/datasets/blastchar/telco-customer-churn

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2. Problem Statement

The objective of this project is to predict whether a telecom customer will churn based on historical customer information. The solution can help businesses identify at-risk customers and develop retention strategies.

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3. Dataset Description

The dataset contains customer information including:

- Demographic details
- Subscription information
- Contract details
- Billing information
- Internet service information
- Churn status

Target Variable:

Churn

- Yes → Customer churned
- No → Customer retained

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4. Exploratory Data Analysis

Several exploratory analyses were performed:

Target Variable Analysis

The distribution of churned and retained customers was analyzed to understand class balance.

Numerical Feature Analysis

Histograms were used to study the distribution of numerical variables such as:

- Tenure
- MonthlyCharges
- TotalCharges

Outlier Detection

Boxplots were created to identify possible outliers.

Correlation Analysis

A correlation heatmap was generated to understand relationships among numerical variables.

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5. Data Preprocessing

The following preprocessing steps were performed:

Missing Value Handling

The TotalCharges column contained 11 missing values which were replaced using the median value.

Feature Encoding

Categorical variables were converted into numerical format using One-Hot Encoding.

Feature Scaling

Numerical features were standardized using StandardScaler.

Train-Test Split

The dataset was divided into:

- Training Set: 80%
- Testing Set: 20%

Random State: 42

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6. Model Development

A Logistic Regression model was implemented using Scikit-Learn.

Hyperparameters

Penalty: L2

Regularization Strength (C): 1.0

Solver: lbfgs

Maximum Iterations: 1000

Justification

- L2 regularization helps reduce overfitting.
- C = 1.0 provides balanced regularization.
- lbfgs is efficient for binary classification.
- Increased iterations ensure model convergence.

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7. Model Evaluation

Classification Report

Metric	Class 0	Class 1
Precision	0.85	0.66
Recall	0.89	0.57
F1-Score	0.87	0.61

Overall Performance

Accuracy: 81%

Macro Average F1-Score: 0.74

Weighted Average F1-Score: 0.80

ROC-AUC Score: 0.84

Confusion Matrix

The confusion matrix was generated to evaluate prediction performance across both classes.

ROC Curve

The ROC curve demonstrated strong discrimination capability with an AUC score of approximately 0.84, indicating good predictive performance.

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8. Interpretation and Insights

The Logistic Regression coefficients were analyzed to understand feature influence.

Key observations include:

- Customers with shorter tenure exhibit higher churn probability.
- Month-to-month contracts are associated with increased churn.
- Customers with stable long-term contracts tend to remain loyal.
- Billing and service usage features significantly influence churn behavior.

These insights can support targeted customer retention programs.

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9. Limitations

- Logistic Regression assumes a linear relationship between features and the log-odds of churn.
- Moderate class imbalance affects recall for churned customers.
- The model may not capture complex non-linear relationships.

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10. Future Work

Potential improvements include:

1. Applying SMOTE for class balancing.
2. Hyperparameter tuning using GridSearchCV.
3. Using Random Forest and XGBoost models.
4. Advanced feature engineering.
5. Ensemble learning approaches.

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11. Conclusion

This project successfully implemented a Logistic Regression model for customer churn prediction. The model achieved an accuracy of 81% and a ROC-AUC score of 0.84, demonstrating good predictive capability. The analysis revealed important business factors influencing churn and provided actionable insights for customer retention strategies.

Overall, Logistic Regression proved to be an effective and interpretable baseline model for telecom customer churn prediction.